Available online at www.sciencedirect.com
ScienceDirect
Agricultural Sciences in China
2009, 8(12): 1448-1457
December 2009

Cloning and Expression of Calcium-Dependent Protein Kinase (CDPK) Gene Family in Common Tobacco (*Nicotiana tabacum*)

TAI Shuai-shuai[1], LIU Guan-shan[1], SUN Yu-he[1] and CHEN Jia[2]

[1] *Tobacco Research Institute, Chinese Academy of Agricultural Sciences/Key Laboratory of Tobacco Quality Control, Ministry of Agriculture, Qingdao 266101, P.R.China*
[2] *State Key Laboratory of Plant Physiology and Biochemistry/College of Biological Sciences, China Agricultural University, Beijing 100193, P.R.China*

Abstract

To further study the function of calcium-dependent protein kinase (CDPK) gene family in common tobacco (*Nicotiana tabacum*), it is necessary to isolate more CDPKs from common tobacco and describe the sequence characteristics, evolutionary relationship and gene expression. Reverse transcription-PCR (RT-PCR), rapid amplification of cDNA (RACE) and bioinformatics methods were used to isolate CDPKs from common tobacco. A phylogenetic tree was created using the MEGA4.0 program and expression patterns of the three full-length CDPK genes were studied by RT-PCR. After all aforementioned efforts, we obtained eight additional common tobacco CDPK genes, of which three possessed complete open reading frames (ORFs). Phylogenetic analysis divided 11 full-length *Nicotiana* CDPK genes into four subfamilies, and two putative common tobacco and *Arabidopsis* orthologous CDPK genes might correspond to well-conserved functions. Three full-length CDPK genes in common tobacco were detected in all tobacco organs tested, but their expression patterns were significantly different. Eight non-redundant common tobacco CDPK genes were isolated in this study. Along with the previously characterized CDPK genes, at least 15 members of the CDPK family exist in common tobacco. This work establishes a foundation for a genome-wide study of this important gene family in common tobacco.

Key words: common tobacco (*Nicotiana tabacum*), CDPK, phylogenetic tree, gene expression

INTRODUCTION

Calcium-dependent protein kinases (CDPKs), identified only in plants and certain protozoans, are a kind of Ser/Thr protein kinases (Cheng *et al*. 2002). CDPKs are directly activated by calcium signal and independent of calmodulin. Typical CDPKs are composed of a polypeptide chain, containing four functional regions (domains), which, listed in order from the amino to carboxyl terminus, are a variable domain, a protein kinase domain, a junction domain, and a regulatory domain (Cheng *et al*. 2002; Harmon *et al*. 2001; Harper *et al*. 1991; Hrabak *et al*. 2003). CDPKs exist widely in roots, stems, leaves, flowers, fruits, and seeds of plants and play important roles in plant calcium signal transduction. Increasing evidence has demonstrated that CDPKs are involved in carbon and nitrogen metabolism, ion and water transmembrane transport, cytoskeleton regulation, stomatal movement regulation and growth and development regulation (Liu and Chen 2003). Meanwhile, CDPKs also play important roles in defense response to antifungal (Romeis *et al*. 2000, 2001) and in injury response to anti-abiotic stress (Chico *et al*. 2002). Expression of

This paper is translated from its Chinese version in *Scientia Agricultura Sinica.*
Correspondence LIU Guan-shan, Associate Professor, Ph D, E-mail: liuguanshan2002@163.com; SUN Yu-he, Professor, Tel: +86-532-88703168, Fax: +86-532-88703168, E-mail: yhsun@163.com

© 2009, CAAS. All rights reserved. Published by Elsevier Ltd.
doi: 10.1016/S1671-2927(08)60358-2

CDPK genes can be elevated by physical and chemical stress, such as low temperature, drought, wounding, salt, nutritional deficiencies, and hormones (Botella *et al.* 1996; Chico *et al.* 2002; Frattini *et al.* 1999; Martínez-Noël *et al.* 2007; Murillo *et al.* 2001; Patharkar and Cushman 2000; Romeis *et al.* 2001; Saijo *et al.* 2000; Sheen 1996; Tsai *et al.* 2007; Urao *et al.* 1994; Wan *et al.* 2007; Yang and Komatsu 2001). Therefore, researches about CDPK genes show very important significance in understanding of plant growth and development as well as in enhancing the capacity of plants resistant to environmental stress. Plant CDPKs represent a very large gene family. It was estimated that there were 34 and 31 CDPK genes in *Arabidopsis thaliana* and rice (*Oryza sativa*) genomes, respectively (Asano *et al.* 2005; Cheng *et al.* 2002; Ray *et al.* 2007). It was assumed that there were 26 CDPK genes in wheat (*Triticum aestivum*) genome at least (Li *et al.* 2008). Since the isolation of the first CDPK cDNA from common tobacco (*Nicotiana tabacum*), only 7 cDNA clones encoding CDPKs have been isolated in common tobacco (Romeis *et al.* 2001; Wang *et al.* 2005; Yoon *et al.* 1999; Zhang *et al.* 2005). In addition, CDPK genes were also cloned from other species in tobacco genus, including one CDPK gene from *Nicotiana benthamiana* (Romeis *et al.* 2001), 4 CDPK genes from *Nicotiana attenuata* (Wu *et al.* 2007) and 2 CDPK genes from *Nicotiana plumbaginifolia*. In the 14 tobacco genus CDPKs, only eight ones have complete open reading frames (ORFs).

Tobacco is not only an important economic leaf crop but also an important model plant. Increasing numbers of tobacco ESTs in public databases and sophisticating of gene cloning technology make it possible to isolate more CDPKs from the important gene family. The objective of the present study was to isolate more CDPK genes from common tobacco and to analyse the sequence characteristics, evolutionary relationship and expression characteristics of tobacco CDPK genes in order to lay foundation for the genome-wide study of functions of the CDPK gene family.

MATERIALS AND METHODS

Plant materials

Flue-cured tobacco variety K326 was cultivated up to reproductive growth stage in greenhouse under nature light. Shoot tips, roots, stems, and leaves during vegetative growth stage and flower buds and flowers during reproductive growth were collected, immediately frozen in liquid nitrogen and maintained at -70°C for RNA extractions.

Cloning of common tobacco CDPK genes

Total RNA was isolated from common tobacco leaf using RNAprep pure Plant Kit (TIANGEN, Beijing, China) according to the manufacturer's instruction and DNase I (TIANGEN, Beijing) was used to avoid genomic DNA contamination. The first-strand cDNA was synthesized with PrimeScript™ RT-PCR Kit (TaKaRa, Dalian, China). Degenerate primers CDPKF and CDPKR (Liu *et al.* 2006) (Table 1) were used to amplify middle CDPK fragments. TaKaRa TP600 gradient PCR amplifiers were used to carry out PCR reaction and reaction conditions consisted of 5 min at 94°C for predenaturing, 30 s at 94°C for denaturing, 30 s at 50°C for annealing and 1 min at 72°C for extension for 35 cycles and a final extension step of 7 min at 72°C. *Ex Taq* HS was purchased from TaKaRa. PCR products were separated on a 0.8% agarose gel, eluted, linked to pGM-T vector (TIANGEN, Beijing) and sequenced in TIANGEN Biochemical Technology Co., Ltd.

5´-RACE and 3´-RACE were performed with SMART™ RACE cDNA Amplification Kit (Clontech, Mountain View, CA, USA). One of the CDPK middle fragments was selected to design gene-specific primers (GSPs) and nest gene-specific primer (NGSPs) (Table 1). 5´-RACE was performed by using 5´GSP and Universal Primer A Mix (UPM) under the following condition: cDNA was denatured at 94°C for 5 min, followed by 5 cycles (94°C for 5 s, 68°C for 10 s and 72°C for 1 min), 5 cycles (94°C for 5 s, 66°C for 10 s and 72°C for 2 min) and 25 cycles (94°C for 5 s, 64°C for 10 s and 72°C for 2 min), and by 10 min at 72°C. The 5´-RACE product was diluted 1 000 times with RNase-free ddH_2O as template for nested PCR reaction, using 5´ NGSP and nested universal primer (NUP) under the following condition: 94°C for 5 min, 30 cycles (94°C for 30 s, 64°C for 30 s and 72°C for 90 s) and 72°C for 10 min. 3´-RACE was performed by using 3´ GSP and UPM under the following condition: 94°C for

© 2009, CAAS. All rights reserved. Published by Elsevier Ltd.

5 min, 30 cycles (94°C for 30 s, 64°C for 30 s and 72°C for 90 s) and 72°C for 10 min. The PCR products were purified, linked to pGEM-T vector and sequenced in TIANGEN Biochemical Technology Co., Ltd.

Based on the spliced full-length cDNA sequence of the 3′- and 5′-RACE products, primers ORFF and ORFR were designed in 5′-UTR and 3′-UTR for the amplification of complete ORF (Table 1). The thermal cycling program was 94°C for 5 min, 35 cycles (94°C for 30 s, 52°C for 30 s and 72°C for 1 min) and 72°C for 10 min. PCR products were purified and linked to pGEM-T vector for sequencing.

In silico cloning of common tobacco CDPK genes

Common tobacco ESTs and stand-alone BLAST program (Altschul *et al*. 1997) were downloaded from NCBI at www.ncbi.nlm.nih.gov and FORMATDB command was used to format all the sequences to construct the local database. *In silico* cloning of common tobacco CDPK genes was based on two strategies. Firstly, homologous sequences were searched against the local tobacco EST database by TBLASTN program, using the amino acid sequence encoded by homologously cloned *NtCDPK5* as a query, then the searched homologous fragments were taken as seed sequences to accomplish the extension and splicing of genes by BLASTN program. Secondly, using the full-length cDNA sequences of 34 *A*. *thaliana* and 31 rice CDPK genes as queries, homologous sequences were searched by BLASTN program and the extension and splicing of genes were accomplished as above. Default parameters were used when BLAST search and false positives were removed manually. ORFs of full-length cDNA sequences were analyzed by ORF finder software at www.ncbi.nlm.nih.gov and gene integrity was verified by homology alignment using BLASTP procedures in the NCBI websites.

Multiple sequence alignment, phylogenetic tree analysis and K_A/K_S estimation

Taking the first common tobacco CDPK protein sequence which was encoded by *NtCDPK1* as the standard sequence, multiple sequence alignments were made among NtCDPK5, NtCDPK6, NtCDPK7, and NtCDPK1 by use of CLUSTALX software (Thompson *et al*. 1997). A neighbor-joining phylogenetic tree (Saitou and Nei 1987) was constructed of 34 *Arabidopsis thaliana* CDPKs, 24 rice CDPKs (24/31, of which seven encoded proteins were not found) and 11 tobacco genus CDPKs by MEGA4.0 software (Tamura *et al*. 2007). According to the phylogenetic tree, K_A/K_S was calculated of orthologous genes between common tobacco and *A*. *thaliana*. K_A (nonsynonymous) and K_S (synonymous) were calculated by the codeml procedure of PAML software package (Yang 1997) at the PAL2NAL online service (http://coot.embl.de/pal2nal/) (Suyama *et al*. 2006).

Expression analysis of common tobacco CDPK genes

Total RNA was extracted from common tobacco stem

Table 1 Primers used in this study

Primer type	Primer name	Sequences
Degenerate primers	NtCDPKF	5′-GG(C/G/T)GG(C/G/T)GA(A/G)(T/C)T(G/T)TT(T/C)GA(T/C)(C/A)G-3′
	NtCDPKR	5′-(C/T)TT(C/T)TT(A/C/T)A(A/G)(C/T)TTATTCAT(G/T)GC-3′
Full-length primers	ORFF	5′-AAAATTGTTTGACTGAACAGTACA-3′
	ORFR	5′-ACATCATACACCTAACCAGAAGC-3′
RACE primers	5′ GSP	5′-GACGCAGAACTTCGGGAGCGACATA-3′
	5′ NGSP	5′-GGCGGCAGCTCGTTCGGAATAGTG-3′
	3′ GSP	5′-CGAACGAGCTGCCGCCGGACTTTG-3′
RT-PCR primers	NtCDPK5F	5′-GTCAGAAGTCAGGCAGTTGGTG-3′
	NtCDPK5R	5′-TTTGCTTTATGAGTTTCGTGGG-3′
	NtCDPK6F	5′-CGAAACGGAATCTACGAACAG-3′
	NtCDPK6R	5′-AAGAAATCAGCAATCACCCTC-3′
	NtCDPK7F	5′-GGGAGTGCTTACTATGTTGCC-3′
	NtCDPK7R	5′-AAGAGTTTGCCTGGCTGTTTG-3′
Internal control primers	NtL25F	5′-ATTAGGCCTGATGGGACGAA-3′
	NtL25R	5′-GGCAACGTCCAAAGCATCAT-3′

© 2009, CAAS. All rights reserved. Published by Elsevier Ltd.

tips, roots, stems, leaves, flower buds, and flowers, and reverse-transcribed to cDNA. Primers were designed according to the gene-specific regions of *NtCDPK5*, *NtCDPK6* and *NtCDPK7* (Table 1). The PCR reaction was performed under the following condition: 94°C for 3 min, 35 cycles (94°C for 10 s, 58/55/57°C for 30 s and 72°C for 1 min) and 72°C for 10 min. Common tobacco ribosomal protein gene *NtL25* (GenBank accession no. L18908) was used as the internal control gene (Table 1) (Volkov *et al*. 2003; Wang *et al*. 2007) to make sure equivalent templates in each PCR reaction. The PCR reaction was performed under the following condition: 94°C for 3 min, 30 cycles (94°C for 10 s, 60°C for 30 s and 72°C for 1 min) and 72°C for 10 min.

RESULTS

Isolation of CDPK genes in common tobacco

Cloning of *NtCDPK5* Using degenerate PCR, two middle fragments of 618 bp were obtained, and a full-length gene was successfully cloned by RACE technology. A band of 750 bp was gained by 5′ RACE; a sequence of 576 bp was acquired by 5′ nested PCR (Fig.1). A sequence of 1 399 bp was gotten by 3′ RACE (Fig.1). After linker sequences in the RACE reactions were removed, a full-length sequence of 1 885 bp, which encoded a protein of 514 amino acids, was assembled according to the overlapping sequences of 5′ and 3′ RACE results. A sequence of 1 767 bp was obtained by full-length sequence amplification (data not shown) and accorded to the assembled sequence. The new common tobacco CDPK gene was named as *NtCDPK5* (*Nicotiana tabacum* calcium-dependent protein kinase 5) and deposited in the GenBank database under accession no. FJ026805. The other 618 bp fragment (the full-length not obtained) was named as *NtCDPK12* and deposited in the GenBank database under accession no. FJ594482.

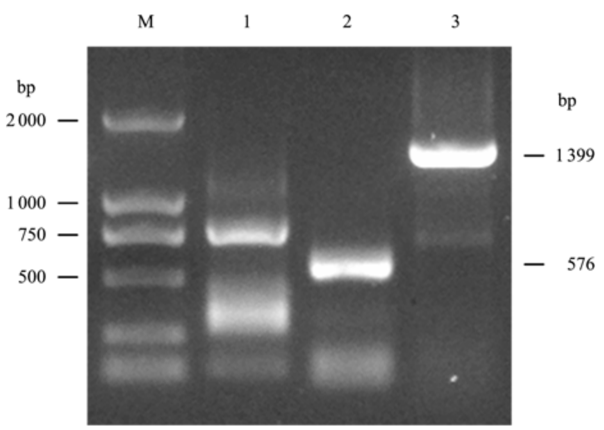

Fig. 1 Products of *NtCDPK5* RACE amplification. M, DL 2 000 marker; lane 1, product of 5′ RACE; lane 2, product of nested 5′ RACE; lane 3, product of 3′ RACE.

***In silico* cloning of six CDPK genes in common tobacco** Using the protein sequence encoded by *NtCDPK5* as a query, four new CDPK genes were obtained (named as *NtCDPK6*, *NtCDPK7*, *NtCDPK8*, and *NtCDPK9*, respectively) from the local tobacco EST database and only two CDPKs had complete ORFs (Table 2). *NtCDPK6* encoded a protein of 530 amino acids and *NtCDPK7* encoded a protein of 521 amino acids. Using full-length cDNA of *A. thaliana* and rice CDPKs as queries, the other two new CDPK genes (named as *NtCDPK10* and *NtCDPK11*, respectively) were acquired but none had full-length sequences (Table 2). Up to now, 15 CDPK genes included the previously reported seven ones were isolated from common tobacco; plus seven CDPK genes from other *Nicotiana* species, 22 CDPK genes were isolated from tobacco genus, of which 11 had complete ORFs (Table 2).

Sequence characteristics of CDPK genes

The deduced amino acid sequences of three full-length CDPK genes isolated in common tobacco in this paper had typical conserved sequences of CDPKs. They had four functional regions (domains), which successively were a variable domain, a protein kinase domain, a junction domain, and a calmodulin-like calcium-binding domain from the amino to carboxyl terminus (Fig.2). There were conserved sequences necessary for myristoylation in the N-terminals of NtCDPK5 and NtCDPK7, whereas there was not such sequence in the N-terminal of NtCDPK6 (Table 2). Variable domains had large changes in sequence and length with little homology. Protein kinase domains

© 2009, CAAS. All rights reserved. Published by Elsevier Ltd.

had typical catalytic conserved sequence for Ser/Thr protein kinases with high homology and no changes in length. Junction domains were composed of 31 amino acids, which were rich of basic amino acids and highly conserved. Calmodulin-like calcium-binding domains had four deduced EF-hand calcium-binding domains and low conservation. These results indicated that the deduced amino acid sequences of these three CDPK genes had classic characteristics of CDPKs and we indeed cloned three complete CDPK genes.

Phylogenetic tree analysis and K_A/K_S value estimation

Similar to *A. thaliana* (Cheng *et al*. 2002) and rice (Asano *et al*. 2005) CDPKs, these 11 complete CDPKs in tobacco genus can be divided into four groups, in which group 2 and group 3 can be again divided into two subgroups, respectively, and group 4 has the least number of CDPKs (Fig.3). NtCDPK5 is located in group 2b, NtCDPK6 in group 3b and NtCDPK7 in group 2a. Common tobacco might have arisen from the hybridization of the diploid species *Nicotiana sylvestris* and *Nicotiana tomentosiformis* (Lee *et al*. 1988). Both proteins encoded by *NtCDPK2* and *NbCDPK2* have 531 amino acids and 97% identity; *NtCDPK2* and *NbCDPK2* were probably the same gene originated from different species genomes. NtCDPK2 and NtCDPK3 have 89% identity and had the same function (Romeis *et al*. 2001); NtCPK4 and NtCPK5 have 86% identity and were located in different subcellular organelles (Wang *et al*. 2005).

Phylogenetic tree analysis indicated that two pairs of orthologous genes *NtCDPK5*/*AtCPK3* and *NtCDPK6*/*AtCPK13* were found in common tobacco and *Arabidopsis thaliana*. The K_A/K_S value of *NtCDPK5*/*AtCPK3* is 0.0292 (K_S: 5.3877; K_A: 0.1574); the K_A/K_S value of *NtCDPK6*/*AtCPK13* is 0.0265 (K_S: 3.0052; K_A: 0.0797). The both K_A/K_S values are far less than one, which indicated that CDPK gene evolution was very conserved and amino acid mutation was conserved under passivated (negative) selection pressure.

Table 2 Characteristics of CDPKs in tobacco genus

Name	Accsession no. (or EST source)	cDNA length (bp)	Amino acids (aa)	EF[1)]	Amino acids at N-terminus	Myristoylation motif[2)]	Subfamilies	References[3)]
NtCDPK1	AF072908	2 251	540	4	MGGCFSKK	Yes	Group 2a	1
NtCDPK2	AJ344154	2 019	581	4	MGNTCVGP	Yes	Group 1	2
NtCDPK3	AJ344155	2 437	578	4	MGNTCVGP	Yes	Group 1	2
NtCPK4	AF435451	2 360	572	4	MGNNCFSS	No	Group 4	3
NtCPK5	AY971376	2 368	567	4	MGSCFSSS	Yes	Group 4	4
NtCDPKL	EF051140	2 022	-	-	-	-	-	-
NtCDPK2	AJ549110	870	-	-	-	-	-	-
NbCDPK2	AJ344156	1 746	581	4	MGNTCVGP	No	Group 1	2
NaCDPK2	EF121304	981	-	-	-	-	-	5
NaCDPK4	EF121310	983	-	-	-	-	-	5
NaCDPK5	EF121305	940	-	-	-	-	-	5
NaCDPK8	EF121306	801	-	-	-	-	-	5
NpCDK8	AJ699160	1 887	528	4	MGNCCVKP	No	Group 3b	-
NpCDK17	AJ699161	1 698	534	4	MGNCCSRG	Yes		-
NtCDPK5	FJ026805	1 885	514	4	MGNCNSSS	Yes	Group 2b	6
NtCDPK6	FG186375, FG187282, FG645577, FG636071,	1 958	530	4	MGNCCRSP	No	Group 3b	6
NtCDPK7	FG141137, FG166066, FG136317, DV160889, FG166139, AM802274	2 152	521	4	MGNICSSK	Yes	Group 2a	6
NtCDPK8	DV162604, AM816929	1 202	-	-	-	-	-	6
NtCDPK9	FG155218, EH618456	1 007	-	-	-	-	-	6
NtCDPK10	FG157177, BP534729	996	-	-	-	-	-	6
NtCDPK11	FG145470	864	-	-	-	-	-	6
NtCDPK12	FJ594482	618	-	-	-	-	-	6

[1)] The numbers of EF hands were predicted by PROSITE scan (Asano *et al*. 2005; Cheng *et al*. 2002).

[2)] N-myristoylation motifs were based on the prediction by PROSITE scan (Asano *et al*. 2005; Cheng *et al*. 2002) with considerations drawn from the references (Romeis *et al*. 2001; Wang *et al*. 2005; Yoon *et al*. 1999; Zhang *et al*. 2005).

[3)] 1, Yoon *et al*. 1999; 2, Romeis *et al*. 2001; 3, Zhang *et al*. 2005; 4, Wang *et al*. 2005; 5, Wang *et al*. 2007; 6, this paper.

© 2009, CAAS. All rights reserved. Published by Elsevier Ltd.

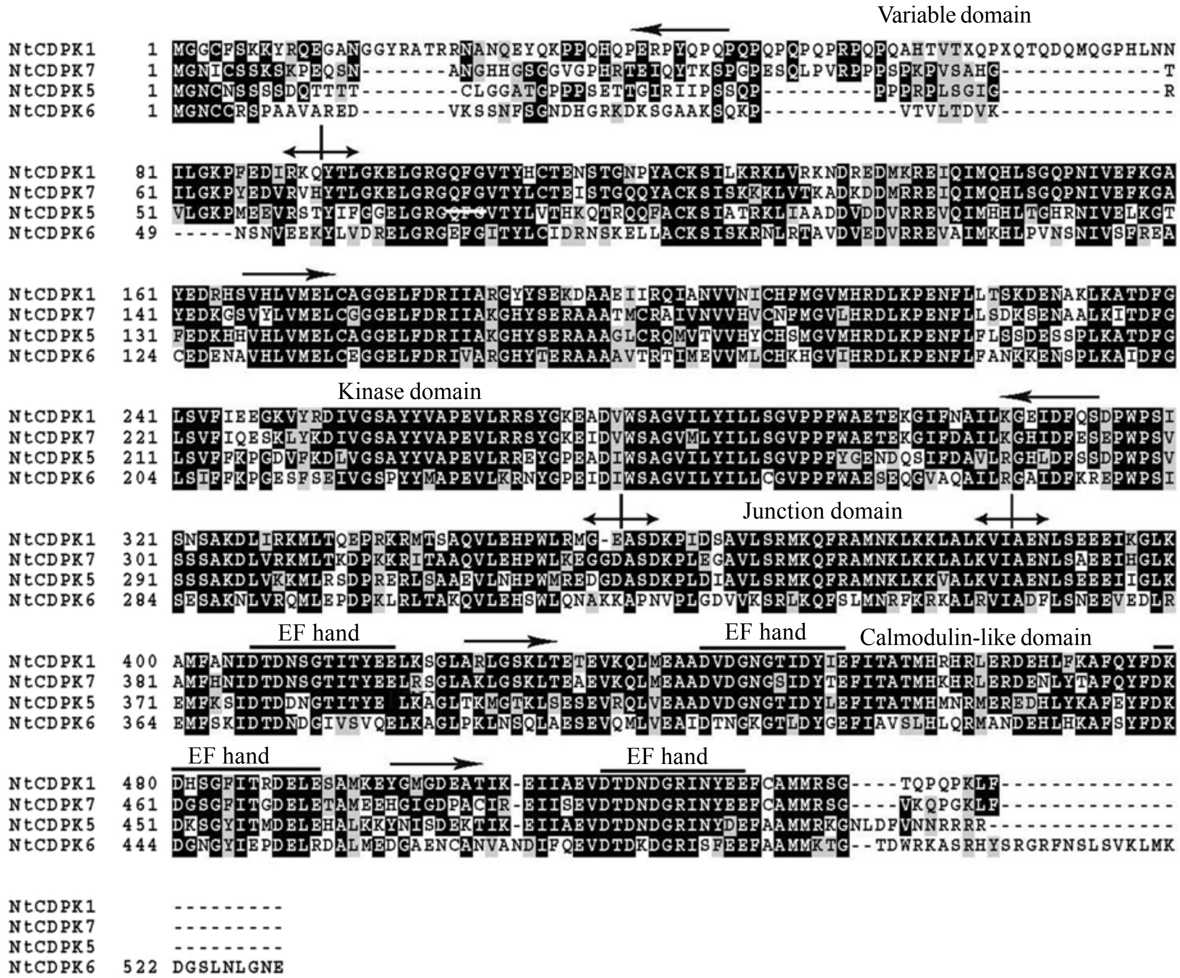

Fig. 2 Multiple sequence alignment of *NtCDPK5, NtCDPK6, NtCDPK7*, and *NtCDPK1* putative amino acid sequences. Identical residues are displayed in black and similar residues are in gray.

Expression analysis of *NtCDPK5*, *NtCDPK6* and *NtCDPK7*

Evidences for CDPK gene expression could be obtained by searching existing EST/cDNA sequences in the databases and more than 90% (20/22) CDPK genes involved in the paper had EST/cDNA evidences (Table 2), of which *NtCDPK5* and *NtCDPK12* sequences were from this study. RT-PCR analysis (Fig.4) showed that the expression of *NtCDPK5* in different tissues had not obvious differences and *NtCDPK5* was a ubiquitously-expressed gene. *NtCDPK6* was highly expressed in shoot tips and flower buds, low expressed in stems and weakly expressed in flowers, leaves and lateral roots. Expression levels of *NtCDPK7* were the highest in leaves, more in lateral roots and stems, less in shoot tips and flowers, and the lowest in flower buds. In a word, these three CDPK genes could be detected in all tissues, but their expressions, were distinctly different among different tissues.

DISCUSSION

So far, only seven CDPK genes were reported in common tobacco, and along with eight CDPK genes obtained in the paper, a total of 15 CDPK genes were separated from common tobacco. Seven CDPK genes were also obtained in several wild species of tobacco genus. CDPKs are a multi-gene family. Compared to *A. thaliana* 34 genes and rice 31 genes, the 15 CDPK

© 2009, CAAS. All rights reserved. Published by Elsevier Ltd.

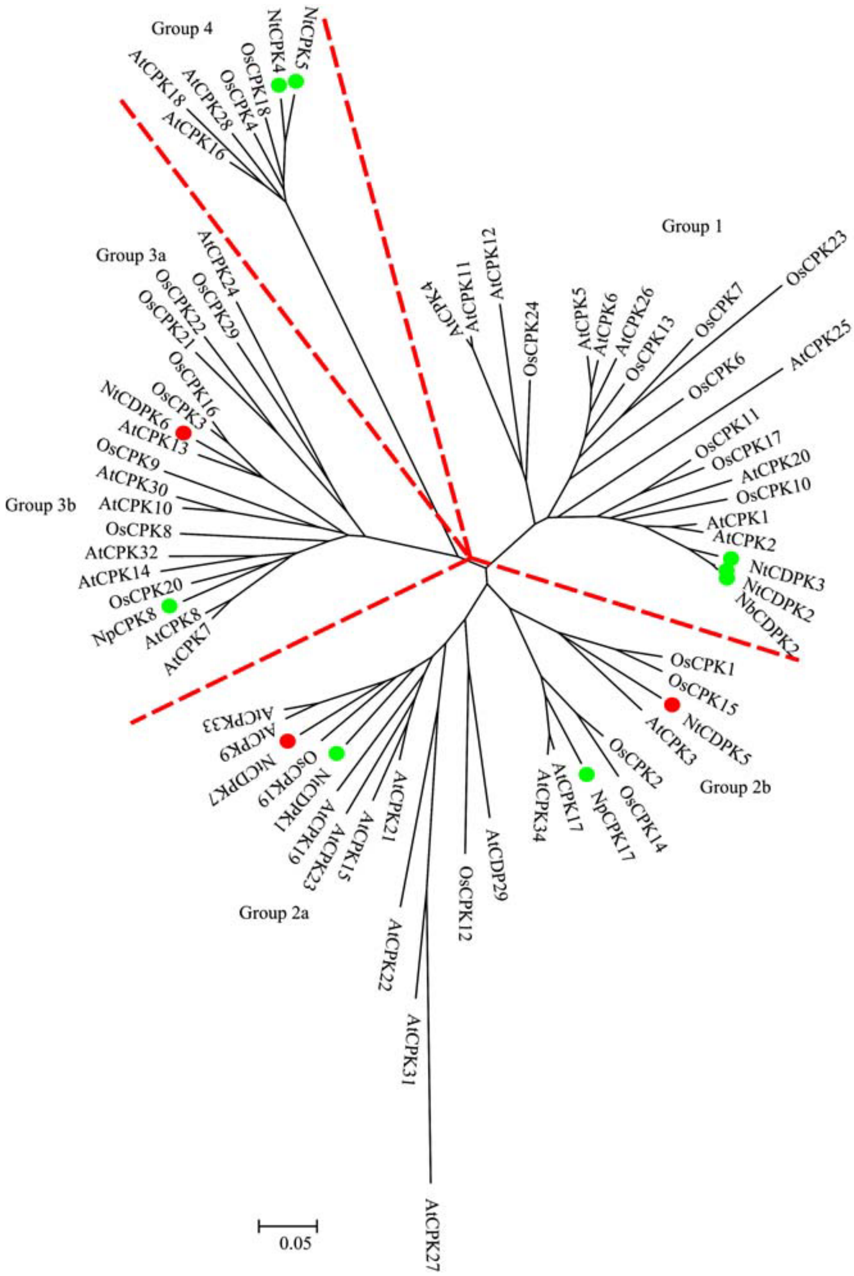

Fig. 3 The neighbor-joining tree of tobacco, *Arabidopsis* and rice CDPK proteins. The full-length common tobacco CDPKs isolated by this study are in red circles; the previously characterized tobacco genus CDPKs are in green circles.

genes may be only one part in common tobacco. Along with the genome database of tobacco to be refined as well as the tobacco genome project that will be started by State Tobacco Monopoly Bureau (China Tobacco Corporation), we believe that there will be more CDPK genes to be isolated. In addition, we found that the *NtCDPKL* gene (GenBank accession no. EF051140) submitted in GenBank was annotated in a partial sequence and the complete ORF could not be found (Table 2). Our amendment showed that there was a lack of an adenine between the 725th and 726th base in the *NtCDPKL* gene. After the base was added (AGGCG A AACTA), the complete ORF could be found; the coding region was located between 34-1 626 bp, encoding 530 amino acids. However, since there were no further experimental evidences, the gene was not involved in the relevant discussion. If *NtCDPKL* is added, 12 complete CDPK genes have been isolated from tobacco

© 2009, CAAS. All rights reserved. Published by Elsevier Ltd.

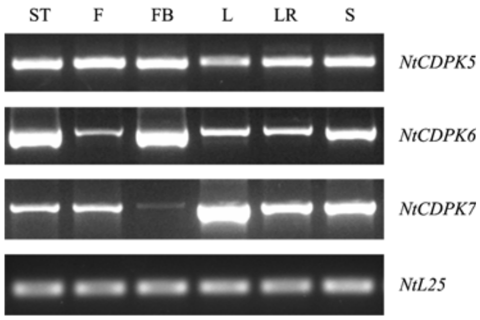

Fig. 4 The expression patterns of *NtCDPK5*, *NtCDPK6* and *NtCDPK7*. Total RNA was isolated from shoot tips (ST), flowers (F), flower buds (FB), leaves (L), lateral roots (LR), and stems (S). Amplification of *NtL25* cDNA was used to ensure that equal amounts of cDNA were added to each PCR reaction.

genus.

Through the construction of the phylogenetic tree, we can analyze the origin relationship among the genes and predict the molecular function. The same subfamily or the same small branch often has the similar function, such as NtCDPK2 and NtCDPK3 on the same branch (Fig.3). The both CDPK transcripts were elevated after race-specific defence elicitation and hypo-osmotic stress and thus participated in the signal transduction pathways induced by different biotic and abiotic stress (Romeis *et al*. 2001). In this paper, *NtCDPK5*/*AtCPK3* and *NtCDPK6*/*AtCPK13* are two pairs of orthologous genes, the K_A/K_S values are far less than one, so their functions may be very conserved. Because AtCPK3 is related to the abscisic acid signal transduction in stomata (Mori *et al*. 2006), we speculate that NtCDPK5 may have the similar function in common tobacco. Along with full-length CDPK genes increasing in tobacco, it is estimated that there will be more homologous genes found, and this will provide a shortcut to further study CDPK gene functions in different species.

Recent studies indicated that spatial expression patterns of CDPK genes in plants could be divided into three categories (Liu *et al*. 2006). The first category was reported to be expressed specifically in given plant tissues. *NtCDPK1* transcripts were present in roots, stems and flowers, but were almost undetectable in leaves (Yoon *et al*. 1999); a rice CDPK gene (spk) was specifically expressed in developing seeds (Kawasaki *et al*. 1993); a maize (*Zea mays* L.) CDPK gene was only detected in the pollen (Estruch *et al*. 1994); the wheat *TaCPK3* was not expressed in developing seeds, and *TaCPK13* was expressed in leaves, young spikes and developing seeds with no traces of transcripts detected in roots or stems (Li *et al*. 2008). The second category did not show significant difference in different tissues. The *Arabidopsis AtCPK3*, *AtCPK8* and *AtCPK12* (Hong *et al*. 1996), and *LeCDPK1* (Chico *et al*. 2002) (*Lycopersicon esculentum*) were ubiquitously expressed in roots, stems, leaves, and flowers. The wheat *TaCPK2* was ubiquitously expressed in roots, stems, leaves, young spikes and developing seeds (Li *et al*. 2008). *NtCDPK5* cloned in this paper was of such genes, which indicated that it might have functions in every organ. The third type could be detected in all tissues, but there was a definite difference in expression. Broad bean (*Vicia faba*) *VfCPK1* was expressed in all tissues tested, but its expression was more abundant in epidermal peels and leaves than in roots, stems and leaves with epidermis removed (Liu *et al*. 2006). Most of the wheat CDPK gene family members had similar expression patterns, such as *TaCDPK5*, 6, 9, 10, 12, 14, 15, and 16 (Li *et al*. 2008). *NtCDPK6* and *NtCDPK7* were also such kind of genes. Through expression analysis, we speculate that NtCDPK6 may play roles mainly in the young parts of common tobacco plants and NtCDPK7 may be mainly involved in vegetative growth.

Further research can use RACE techniques or screen full-length cDNA library to get more full-length cDNA of common tobacco CDPKs. Based on the analysis of the phylogenetic tree, after selecting representative CDPK genes from four sub-families and combining with the research findings in *Arabidopsis*, rice and other species, real-time PCR and RNAi technology can be used to comprehensively study the functions of CDPK gene family in common tobacco.

CONCLUSION

In this study, three full-length CDPK genes and five CDPK gene fragments were cloned in common tobacco by RACE technology and bioinformatics, and the expression profile of three full-length genes in different common tobacco tissues was studied by RT-PCR. Phylogenetic tree analysis showed that 11 full-length CDPK genes in tobacco genus were located in four sub-

© 2009, CAAS. All rights reserved. Published by Elsevier Ltd.

families, and two pairs of orthologous genes were predicted in common tobacco and *Arabidopsis thaliana*.

Acknowledgements

This study was supported in part by the Key Scientific and Technological Development Project from State Tobacco Monopoly Bureau, China (110200701021, 110200701022) and the open subject from State Key Laboratory of Plant Physiology and Biochemistry, China (PPB08004).

References

Altschul S F, Madden T L, Schäffer A A, Zhang J, Zhang Z, Miller W, Lipman D J. 1997. Gapped BLAST and PSI-BLAST: a new generation of protein database search programs. *Nucleic Acids Research*, **25**, 3389-3402.

Asano T, Tanaka N, Yang G X, Hayashi N, Komatsu S. 2005. Genome-wide identification of the rice calcium-dependent protein kinase and its closely related kinase gene families: comprehensive analysis of the *CDPKs* gene family in rice. *Plant Cell Physiology*, **46**, 356-366.

Botella J R, Arteca J M, Somodevilla M, Arteca R N. 1996. Calcium-dependent protein kinase gene expression in response to physical and chemical stimuli in mungbean (*Vigna radiata*). *Plant Molecular Biology*, **30**, 1129-1137.

Cheng S H, Willmann M R, Chen H C, Sheen J. 2002. Calcium signaling through protein kinases. The *Arabidopsis* calcium-dependent protein kinase gene family. *Plant Physiology*, **129**, 469-485.

Chico J M, Raices M, Tellez-Inon M T, Ulloa RM. 2002. A calcium-dependent protein kinase is systemically induced upon wounding in tomato plants. *Plant Physiology*, **128**, 256-270.

Estruch J J, Kadwell S, Merlin E, Crossland L. 1994. Cloning and characterization of a maize pollen-specific calcium-dependent calmodulin-independent protein kinase. *Proceedings of the National Academy of Sciences of the USA*, **91**, 8837-8841.

Frattini M, Morello L, Breviario D. 1999. Rice calcium-dependent protein kinase isoforms OsCDPK2 and OsCDPK11 show different responses to light and different expression patterns during seed development. *Plant Molecular Biology*, **41**, 753-764.

Harmon A C, Gribskov M, Gubrium E, Harper J F. 2001. The CDPK superfamily of protein kinases. *New Phytologist*, **151**, 175-183.

Harper J F, Sussman M R, Schaller G E, Putnam-Evans C, Charbonneau H, Harmon A C. 1991. A calcium-dependent protein kinase with a regulatory domain similar to calmodulin. *Science*, **252**, 951-954.

Hong Y, Takano M, Liu C M, Gasch A, Chye M L, Chua N H. 1996. Expression of three members of the calcium-dependent protein kinase gene family in *Arabidopsis thaliana*. *Plant Molecular Biology*, **30**, 1259-1275.

Hrabak E M, Chan C W M, Gribskov M, Harper J F, Choi J H, Halford N, Kudla J, Luan S, Nimmo H G, Sussman M R, Thomas M,Walker S K, Zhu J K, Harmon A C. 2003. The *Arabidopsis* CDPK-SnRK superfamily of protein kinases. *Plant Physiology*, **132**, 666-680.

Kawasaki T, Hayashida N, Baba T, Shinozaki K, Shimada H. 1993. The gene encoding a calcium-dependent protein kinase located near the *sbe1* gene encoding starch branching enzyme I is specifically expressed in developing rice seeds. *Gene*, **129**, 183-189.

Lee K Y, Townsend J, Tepperman J, Black M, Chui C F, Mazur B, Dunsmuir P, Bedbrook J. 1988. The molecular basis of sulfonylurea herbicide resistance in tobacco. *The EMBO Journal*, **7**, 1241-1248.

Li A L, Zhu Y F, Tan X M, Wang X, Wei B, Guo H Z, Zhang Z L, Chen X B, Zhao G Y, Kong X Y, Jia J Z, Mao L. 2008. Evolutionary and functional study of the CDPK gene family in wheat (*Triticum aestivum* L.). *Plant Molecular Biology*, **66**, 429-443.

Liu G S, Chen J. 2003. Roles of calcium-dependent protein kinases (CDPKs) in plant calcium signal transduction. *Chinese Bulletin of Botany*, **20**, 160-167. (in Chinese)

Liu G S, Chen J, Wang X C. 2006. *VfCPK1*, a gene encoding calcium-dependent protein kinase from *Vicia faba*, is induced by drought and abscisic acid. *Plant, Cell and Environment*, **29**, 2091-2099.

Martínez-Noël G, Nagaraj V J, Calo G, Wiemken A, Pontis H G. 2007. Sucrose regulated expression of a Ca^{2+}-dependent protein kinase (TaCDPK1) gene in excised leaves of wheat. *Plant Physiology and Biochemistry*, **45**, 410-419.

Mori I C, Murata Y, Yang Y, Munemasa S, Wang Y F, Andreoli S, Tiriac H, Alonso J M, Harper J F, Ecker J R, Kwak J M, Schroeder J I. 2006. CDPKs CPK6 and CPK3 function in ABA regulation of guard cells-type anion- and Ca^{2+}-permeable channels and stomatal closure. *PLoS Biology*, **4**, 1749-1762.

Murillo I, Jaeck E, Cordero M J, San Segundo B. 2001. Transcriptional activation of a maize calcium-dependent protein kinase gene in response to fungal elicitors and infection. *Plant Molecular Biology*, **45**, 145-158.

Patharkar O R, Cushman J C. 2000. A stress-induced calcium-dependent protein kinase from *Mesembryanthemum crystallinum* phosphyorylates a two-component pseudo-response regulator. *The Plant Journal*, **24**, 679-691.

Ray S, Agarwal P, Arora R, Kapoor S, Tyagi A K. 2007. Expression analysis of calcium-dependent protein kinase gene

© 2009, CAAS. All rights reserved. Published by Elsevier Ltd.

family during reproductive development and abiotic stress conditions in rice (*Oryza sativa* L. ssp. *indica*). *Molecular and General Genetics*, **278**, 493-505.

Romeis T, Ludwig A A, Martin R, Jones J D G. 2001. Calcium-dependent protein kinases play an essential role in a plant defense response. *The EMBO Journal*, **20**, 5556-5567.

Romeis T, Piedras P, Jones J D G. 2000. Resistance gene-dependent activation of a calcium-dependent protein kinase in the plant defense response. *The Plant Cell*, **12**, 803-815.

Saijo Y, Hata S, Kyozuka J, Shimamoto K, Izui K. 2000. Over-expression of a single Ca^{2+}-dependent protein kinase confers both cold and salt/drought tolerance on rice plants. *The Plant Journal*, **23**, 319-327.

Saitou N, Nei M. 1987. The neighbor-joining method: a new method for reconstructing phylogenetic trees. *Molecular Biology and Evolution*, **4**, 406-425.

Sheen J. 1996. Ca^{2+}-dependent protein kinases and stress signal transduction in plants. *Science*, **274**, 1900-1905.

Suyama M, Torrents D, Bork P. 2006. PAL2NAL: robust conversion of protein sequence alignments into the corresponding codon alignments. *Nucleic Acids Research*, **34** (Web Server issue), W609-612.

Tamura K, Dudley J, Nei M, Kumar S. 2007. MEGA4: molecular evolutionary genetics analysis (MEGA) software version 4.0. *Molecular Biology and Evolution*, **24**, 1596-1599.

Thompson J D, Gibson T J, Plewniak F, Jeanmougin F, Higgins D G. 1997. The CLUSTAL_X windows interface: flexible strategies for multiple sequence alignment aided by quality analysis tools. *Nucleic Acids Research*, **25**, 4876-4882.

Tsai T M, Chen Y R, Kao T W, Tsay W S, Wu C P, Huang D D, Chen W H, Chang C C, Huang H J. 2007. *PaCDPK1*, a gene encoding calcium-dependent protein kinase from orchid, *Phalaenopsis amabilis*, is induced by cold, wounding, and pathogen challenge. *Plant Cell Reports*, **26**, 1899-1908.

Urao T, Katagiri T, Mizoguchi T, Yamaguchi S K, Hayashida N, Shinozaki K. 1994. Two genes that encode Ca^{2+}-dependent protein kinases are induced by drought and high-salt stresses in *Arabidopsis thaliana*. *Molecular and General Genetics*, **244**, 331-340.

Volkov R A, Panchuk I I, Schoffl F. 2003. Heat-stress-dependency and developmental modulation of gene expression: the potential of house-keeping genes as internal standards in mRNA expression profiling using real-time RT-PCR. *Journal of Experimental Botany*, **54**, 2343-2349.

Wan B, Lin Y, Mou T. 2007. Expression of rice Ca^{2+}-dependent protein kinases (CDPKs) genes under different environmental stresses. *FEBS Letters*, **581**, 1179-1189.

Wang X, Replogle A, Davis E L, Mitchum M G. 2007. The tobacco *Cel7* gene promoter is auxin-responsive and locally induced in nematode feeding sites of heterologous plants. *Molecular Plant Pathology*, **8**, 423-436.

Wang Y, Zhang M, Ke K, Lu Y T. 2005. Cellular localization and biochemical characterization of a novel calcium dependent protein kinase from tobacco. *Cell Research*, **15**, 604-612.

Wu J Q, Hettenhausen C, Meldau S, Baldwin L T. 2007. Herbivory rapidly activates MAPK signaling in attacked and unattacked leaf regions but not between leaves of *Nicotiana attenuata*. *The Plant Cell*, **19**, 1096-1122.

Yang G, Komatsu S. 2001. Involvement of calcium-dependent protein kinase in rice (*Oryza sativa* L.) lamina inclination caused by brassinolide. *Plant Cell Physiology*, **41**, 1243-1250.

Yang Z H. 1997. PAML: a program package for phylogenetic analysis by maximum likelihood. *Computer Applications in the Biosciences*, **13**, 555-556.

Yoon G M, Cho H S, Ha H J, Liu J R, Lee H P. 1999. Characterization of *NtCDPK1*, a calcium-dependent protein kinase gene in *Nicotiana tabacum*, and the activity of its encoded protein. *Plant Molecular Biology*, **39**, 991-1001.

Zhang M, Liang S, Lu Y T. 2005. Cloning and functional characterization of *NtCPK4,* a new tobacco calcium-dependent protein kinase. *Biochimica et Biophysica Acta*, **1729**, 174-185.

(Edited by WANG Ning)

© 2009, CAAS. All rights reserved. Published by Elsevier Ltd.